

Anchor Paper

Adam Vinestock  Omer Wachman 

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1 Selected Model Scheme

The paper *Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources* outlines two primary processes for prediction given a user session. Initially, an ensemble of models is employed for candidate prediction, followed by the utilization of a LightGBM ranker for the final prediction. Additionally, various approaches for handling the training data, specifically the User Rating Matrix (URM), with respect to temporal decay are explored. We selected the URM W variant which incorporates temporal weight of sessions by distance (measured in days) from the test sessions. The training data spans across 17 months, a timeframe extensive enough to allow change in users preferences and shifts in fashion trends. Sessions are defined as the daily activities of users, we observed that over 16% of sessions are split across the day (we considered a split session with more than 30 minutes between two consecutive views). Due to these inconsistencies, we chose to discard temporal decay within sessions (URM WW). We opted for GRU as the deep RNN model. This decision was mainly based on its superior performance compared to other models within the respective group. Additionally the sequential nature of the task lead us to chose one of the RNN deep models as oppose to the transformer based model. For the LightGBM ranker models we decreased the number of estimators from 2000 to 100 to reach reasonable runtimes.

2 Setup

This project is hosted on Google Drive and leverages Google Colab notebooks for execution, utilizing its free resources (T4 GPU for deep models). The notebooks can be directly viewed, or downloaded and executed locally with the original dataset.

3 Technical Challenges

Implementing the original code posed significant challenges due to setup and version compatibility issues. Initially, we established a GitHub repository to facilitate both public access to the project and a collaborative working environment. However, we encountered immediate challenges as the size of the training and processed data exceeded the capacity for storage on GitHub. Consequently, we transitioned to conducting the project within Google Drive. The entire project directory was transferred to Google Drive, where the code is executed using the free resource of Google Colab notebooks. While this setup provided a viable solution, it also introduced limitations such as the pipeline's requirement for 9GB of free space, which is substantial for a free Google Drive account. Another challenge we encountered was the lack of documentation and challenging readability of the original writers code.

4 EDA Insights

Our exploratory data analysis involved several key steps:

- **Dataset Characteristics Overview:** We examined the dataset's fundamental attributes, including data types, dimensions, presence of missing values, duplicates, and the uniqueness of item IDs across the training, test, and candidate sets. The dataset is divided into a training set of 1 million sessions and a test set of 100k sessions. The test set is further split into two subsets of 50k sessions for validation and final testing. Our analysis revealed no missing values or duplicates. However, we noted the presence of 73 items in the test set and 62 items in the candidate set that were not included in the training set. This highlights the necessity to account for unseen items during training, particularly when making predictions on test data.
- **Temporal Trends in Purchases:** Analysis of daily purchase frequencies indicated a consistent pattern of sessions over time. Notably, there were identifiable upticks on special dates, aligning with major events such as Halloween (October 31st), Thanksgiving (November 26th), and Black Friday (November 27th).
- **Session Duration Analysis:** We investigated the length and dynamics of user sessions. The average session was observed to have 5.7 views including purchase with an average time of 658 seconds between views, and a median duration of 47 seconds. Interestingly, the average time interval from the last view to a purchase was 873 seconds, which is marginally longer than the typical time between views maybe due to transaction actions. Each session ID represents a user's activity over a day. Notably, in over 16.2% of sessions, a time gap exceeding 30 minutes was observed, indicating occasional prolonged user engagement or breaks within sessions.
- **Diversity in Item Features:** The analysis revealed a significant variation in the distribution of categories and feature values. For instance, category 27 contains only one item, whereas category 47 has over 23,000 items. A similar pattern is observed in feature values. The visual representation of item count by category and value allows to visually traverse along the X and Y lines and view variations in distribution along with items count. This diversity suggests a spectrum in item features, ranging from widespread, generic attributes to more specialized, unique ones.
- **Selecting item subgroup by feature values:** Our innovation includes boosting performance of an item subgroup, this step was performed to get a sense of how many feature values need to be selected randomly for slicing a substantial section of the items. Here we observe that to reach a 10% slice of the items 107 feature values needed to be selected.
- **Stratification by Feature Values:** Our innovation includes boosting performance of an item subgroup. To achieve this, we want to emulate performing stratification of items by feature values and analyze whether the proportions are consistent across the entire item set, item appearance in training, and item appearance in test set. We randomly sampled item IDs to represent approximately 10% of the total item pool. In analyzing five such samples, we found that these selections accounted for an average of 12.78% of all item IDs. When considering occurrences in the training data, these represented 11.45% of item IDs, whereas in the test data, the figure was 8.42%. This discrepancy indicates that to achieve a significant representation in the test data, the initial subgroup size needs to be larger.

5 Results

The following table presents a comparative analysis of MRR@100 scores across various parts of the ensemble model in our configuration. All models used URM_W as the user rating matrix, where 'ALL' includes UCF, ICF, ICBF and GRAPH as candidate selectors. The setups used 10 lightGBM rankers with 100 estimators after candidate selection.

When comparing results to those reported in the original study, we can see lower MRR scores with our models. The most comparable value of which is the '10 Rankers with C1,N1', where they achieved a score of 0.1970 vs our 0.1856. This discrepancy can be attributed to our methodology, which involved the exclusive use of a single neural network architecture (GRU), a single user rating matrix and reduced estimators in the LightGBM rankers.

	UCF,GRU	ICF,GRU	ICBF,GRU	GRAPH,GRU	ALL,GRU
MRR@100 Training	0.1742	0.1788	0.1748	0.1818	0.1880
MRR@100 Leaderboard	0.1879	0.1897	0.1694	0.1884	0.1856

Table 1: Quantitative measure of each recommendation model

Appendix

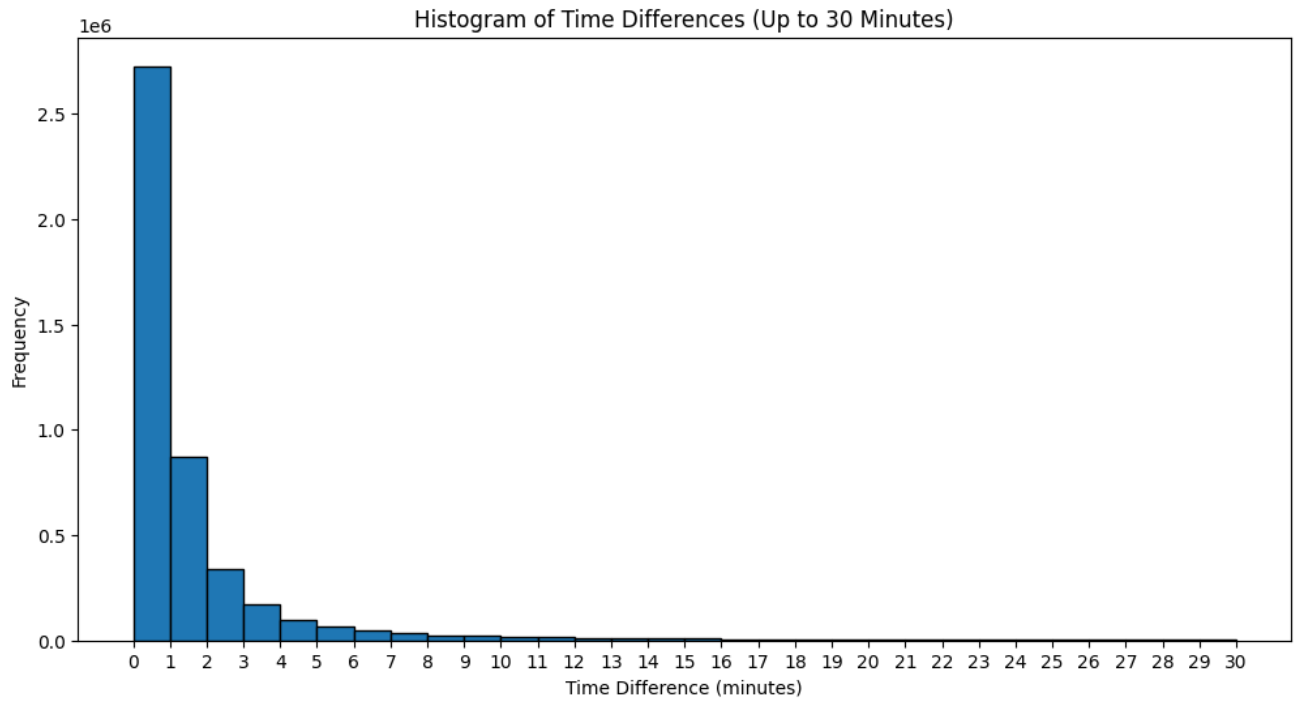


Figure 1: Session duration analysis, count of time between views including purchase, additional 206015 time differences were observed above 30 minutes.

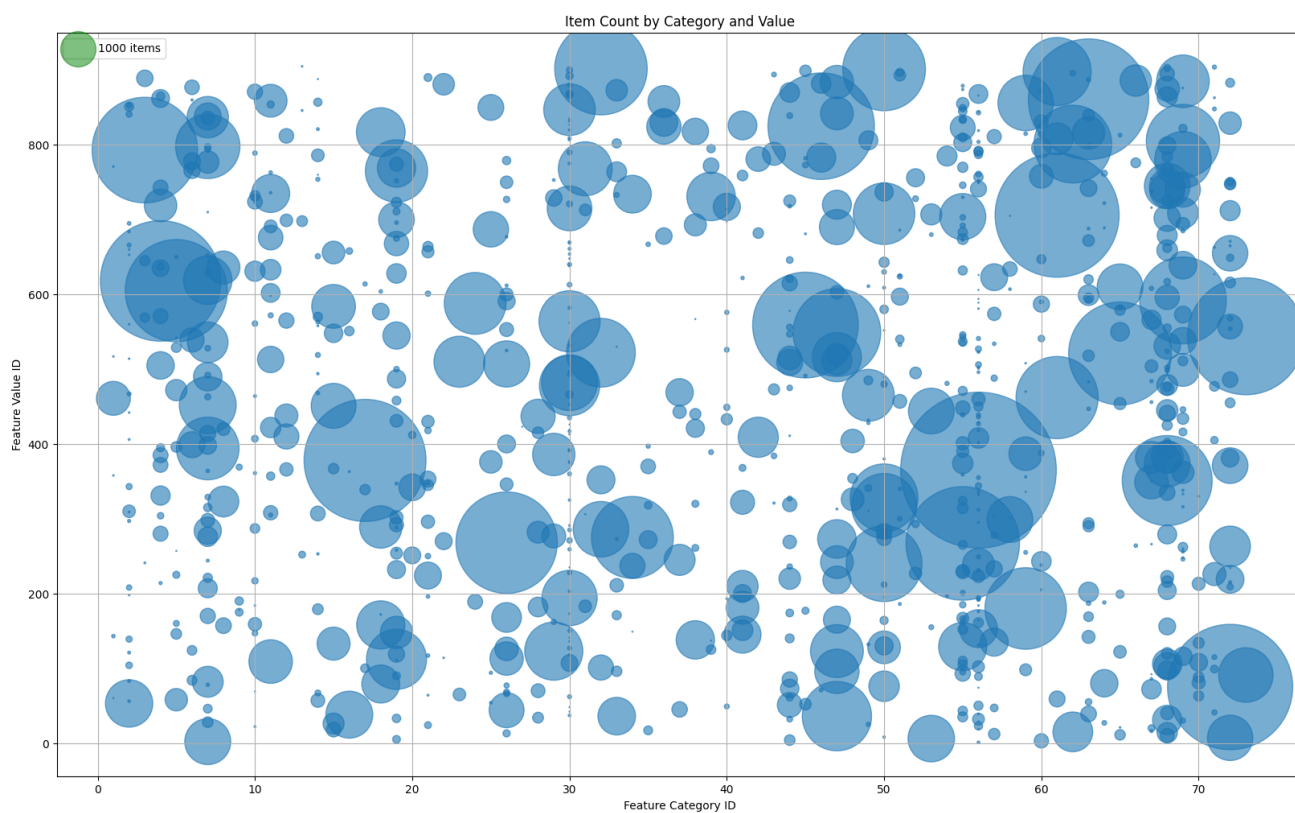


Figure 2: Analyzing diversity in item features, a count of items is displayed for each category-feature value.